Does the between-arm SAE contrast survive a second dictionary?

Findings 19 and 21 compare a model that must deduce against one that reads the adjacency-order leak. Both arms were one dictionary each until tonight.

YES AT THE LAYER IT IS QUOTED AT: +0.496 -> +0.487 at L8.

THE QUOTED NUMBER REPLICATES. At L8, best justifies_next F1 at pre_color is 0.743 (twin) against 0.247 (leak-reader) at seed 0, and 0.721 against 0.233 at seed 1. The gap moves by 0.009. justifies_next[u] means 'u is a KNOWN neighbour of the node being emitted' -- the input to DD's elimination rule.

BUT MY PRE-REGISTRATION ASKED FOR THE WRONG STATISTIC, AND IT FAILS AT BOTH SEEDS. It said 'max over layers' for both arms and then quoted a seed-0 baseline of 0.496 that is the twin's MAX against the leak-reader's value AT L8. Scored as written: +0.233 at seed 0 and +0.202 at seed 1, both under the 0.30 bar. The arithmetic was wrong before the job ran, so this is a scoring error, not a replication failure.

WHICH MAKES THE RESULT SHARPER, NOT WEAKER. The leak-reader's forcing-set peak is at L10 (0.519), the twin's at L8 (0.721). The difference between the two models is WHEN the forcing set appears, not whether: the twin has it by L8, inside the L6-L10 window where activation patching shows the causal fact being handed over; the leak-reader only gets one at L10, after that window has closed.

AND THE ASYMMETRY IS NOW 6 DICTIONARIES DEEP. justifies_next features at pre_node -- the positions where the model picks the NODE rather than its COLOUR -- total 0 across all 42 layer-measurements ever made: seven layers x three checkpoints (solver, twin, leak-reader) x two seeds.

Job 50604600, 57 min on one H100, COMPLETED 2026-10-31 23:31. --seed 1 is the only change from job 49744985, the run findings 19 and 21 were written from. Both arms re-fit in the same job; every number on this deck is read from the two runs' CSVs.

Were the two dictionaries fit under the same pressure?

If seed 1's fits were looser, a smaller gap would mean nothing. They are not.

layer	l1 twin s0	l1 twin s1	l1 leak s0	l1 leak s1	var expl twin	var expl leak	L0 twin	L0 leak
L0	**0.0000**	**0.0000**	**0.0000**	**0.0000**	**1.0000/1.0000**	**1.0000/1.0000**	**5.9/8.2**	**7.0/7.9**
L2	0.2000	0.2000	0.2000	0.2000	0.9907/0.9906	0.9909/0.9904	64.4/64.8	67.6/68.4
L4	0.2000	0.2000	0.2000	0.2000	0.9891/0.9891	0.9899/0.9897	55.3/55.3	64.6/65.2
L6	0.2000	0.2000	0.2000	0.2000	0.9880/0.9880	0.9891/0.9888	60.8/61.3	56.8/57.7
L8	0.4000	0.4000	0.2828	0.2828	0.9749/0.9749	0.9827/0.9824	56.9/56.8	59.1/59.3
L10	0.4000	0.4000	0.4000	0.4000	0.9717/0.9720	0.9731/0.9730	61.4/61.6	63.9/64.0
L12	0.4000	0.4000	0.4000	0.4000	0.9718/0.9713	0.9758/0.9727	71.6/71.4	60.3/59.1

l1_final is IDENTICAL across seeds at every layer of both arms from L2 up -- the l1 search lands on the same coefficient each time. The one exception is the twin's layer 0, where the search bottoms out near zero (2e-05 against 6e-06, both printing as 0.0000). Variance explained agrees to 4 decimals and achieved sparsity to about 1 in 60 against a target of 64. LAYER 0 (red) is degenerate in all four fits, exactly as it was in the solver (finding 37): L0 collapses to 6-8 against the target of 64 with ~3932 of 4096 features dead, so its F1 is stable and its feature COUNTS are not. That is a property of the layer, not of a seed.

ONE CODE DELTA, AND IT IS ADDITIVE.

spin_gnn/sudoku_sae.py changed once between the seed-0 arms (10-27) and tonight: commit 10c38b0 (10-28) added two WRITES -- the per-dimension normalisation stats the ablation work needs, and the per-kind per-feature CSVs. Nothing in the fitting or scoring path moved, which the identical l1_final values above independently confirm. Every other flag is byte-identical: expansion 16, 1500 SAE epochs, 3000 instances, 40 positions each, alive-rate 1e-3, 19 null shifts, target L0 64, --split-by-kind.

--seed moves four things together in sudoku_sae: the instance pool, which positions are sampled, the SAE initialisation and the null-shift RNG. So this is a full replication, not an init-only ablation -- agreement is strong, and a disagreement could not have been pinned on any one of the four.

What the model actually sees

Every token is one categorical id into a 34-symbol vocabulary (N=24, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING
	BEGIN_ROUND SOLVED UNRESOLVED
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3
	the q colours (token name COLOR_n, prose 'colour')
10-33	N0 N1 ... N23
	one id per node, no structure in the id

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*
SOLVED	

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 198 TOKENS

HINTS	BEGIN_HINTS N2 COLOR_2 N3 COLOR_2 N5 COLOR_2 N8 COLOR_3 N10 COLOR_0 N11 COLOR_1 ... (31 tokens)
GRAPH	BEGIN_GRAPH N0 N16 N5 N18 N13 N14 N1 N8 N13 N2 N11 N19 ... (145 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N0 COLOR_1 N1 COLOR_0 N4 COLOR_0 N6 COLOR_1 N9 COLOR_1 N12 ... (22 tokens)

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 2 rounds, 9 deductions,
18 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000004.

One real reasoning trace

15 hints of 24 nodes, then 2 rounds of simultaneous deduction. Hints: N2=COLOR_2 N3=COLOR_2 N5=COLOR_2 N8=COLOR_3 N10=COLOR_0 N11=COLOR_1 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND            N0 COLOR_1 N1 COLOR_0 N4 COLOR_0 N6 COLOR_1 ...
  8 deduced
BEGIN_ROUND            N7 COLOR_2
  1 deduced
SOLVED
```

WHY ONE DEDUCTION FIRES

node N0, neighbours N16 N5 N18 N13 N14

their colours known before this round:

```
   [0, 2, 3]  (3 = q-1 of the 4)

   (4 known neighbours, 3 DISTINCT colours -- some repeat)

   still unknown: N16
```

only colour left: 1

```
so the trace emits  N0 COLOR_1
```

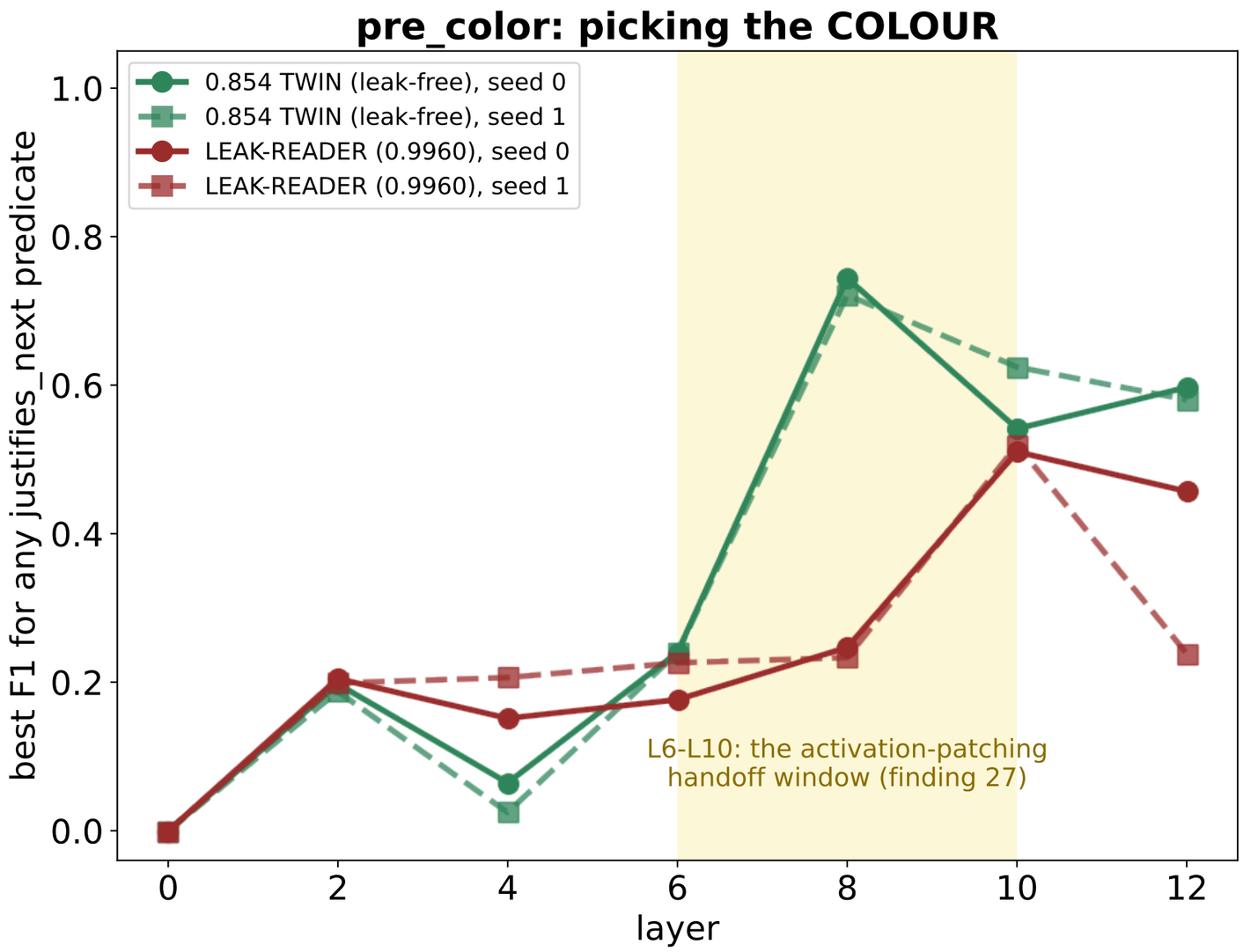
The model is never shown this reasoning -- only the node/colour pair.

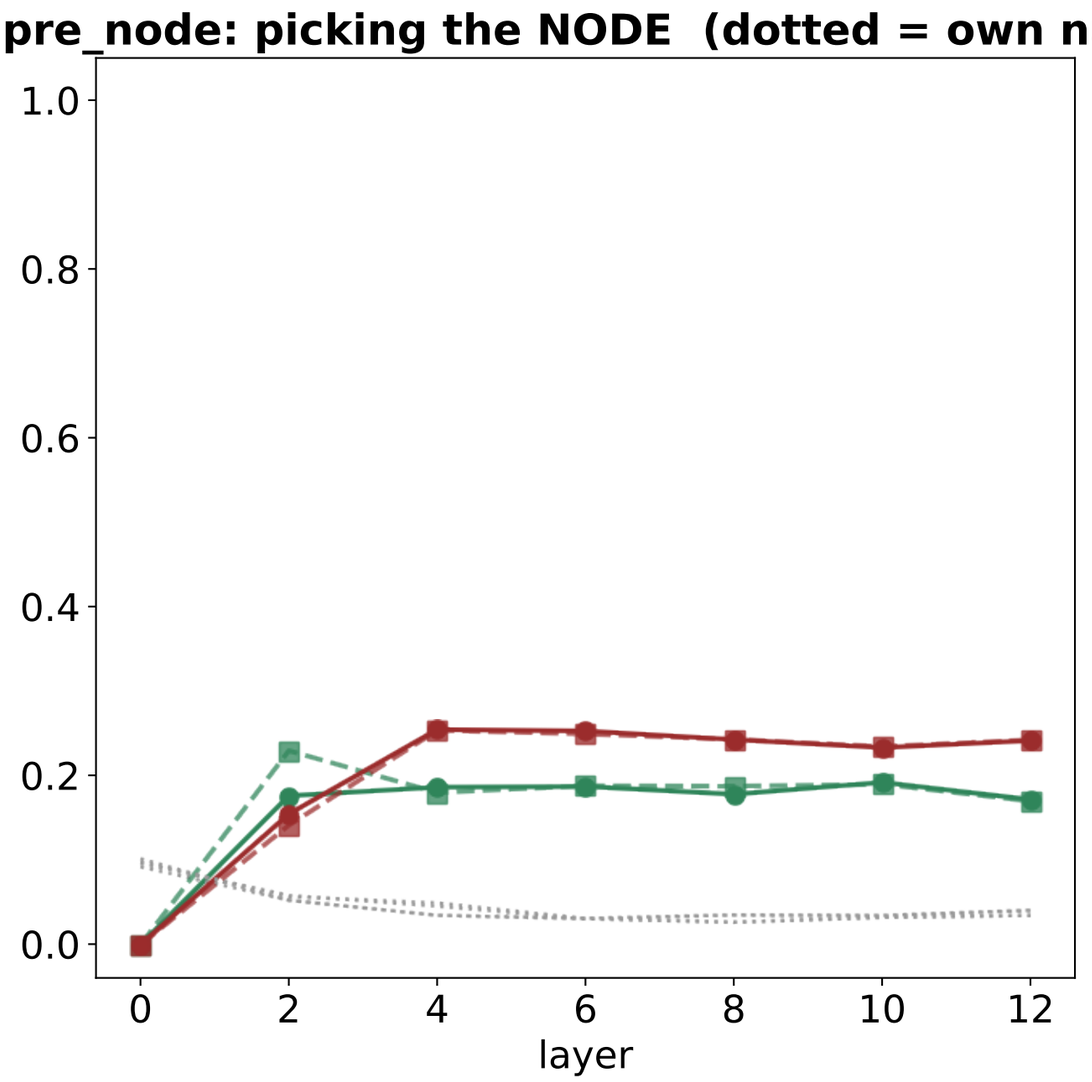
A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

The SAE is fit on the model's activations at exactly these positions. pre_node = the step before it emits a NODE id, pre_color = the step before it emits that node's COLOUR. justifies_next[u] is true when u is one of the already-known neighbours whose colours force the answer -- the list shown on the right of this page, for N0.

T=2 rounds, the median of 40 draws from this config (range 1-6) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

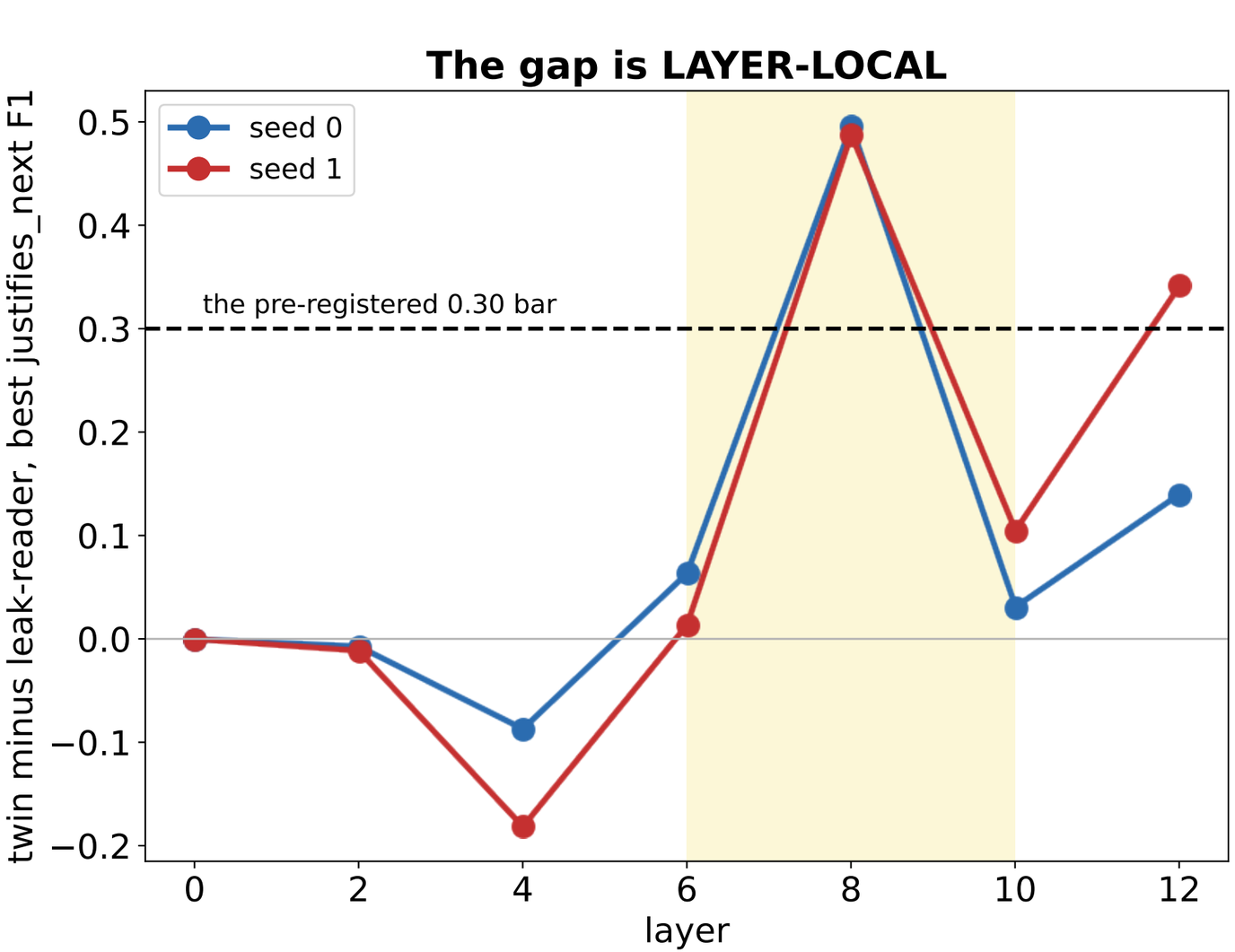
Four dictionaries, one shape: nothing at pre_node, a back-half bump at pre_color

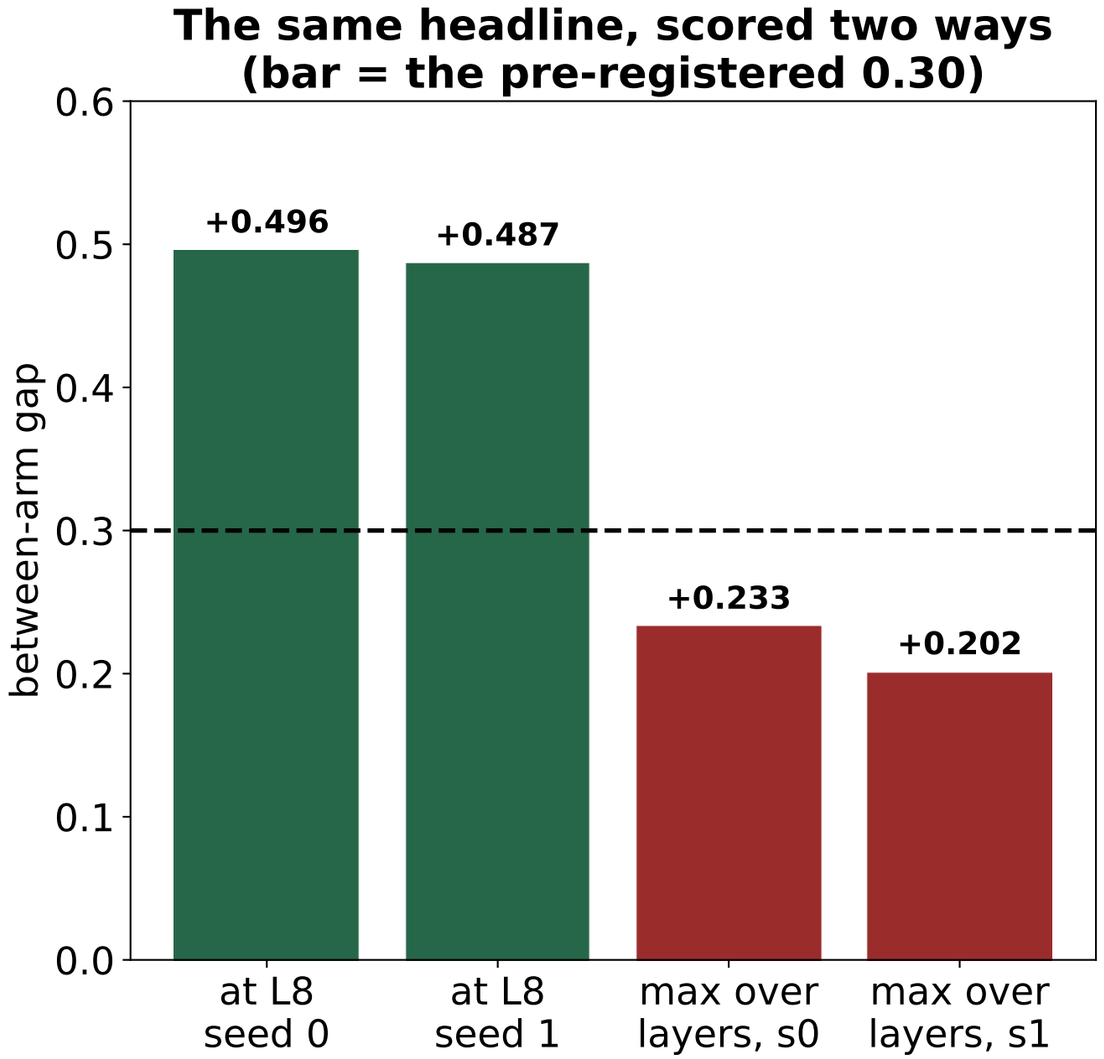




Right panel: every curve sits at or under its own shuffled null, and the feature COUNT is zero at all seven layers in all four dictionaries -- the asymmetry findings 21 and 29 rest on. Left panel: the two arms are far apart through L8 and converge at L10, which is the whole story of this run.

Where the two models differ is WHEN the forcing set shows up





Both scorings are honest and they disagree, so the write-up has to say which it means. The project quotes L8 (findings 19, 21, 29), where the gap is ~0.49 in both seeds; max-over-layers is ~0.21 because the leak-reader catches up at L10. The pre-registration mixed the two and set its bar from the mixture.

The four pre-registrations, scored

Written into scripts/slurm/sudoku2_sae_kindsplit_seed1.sbatch before submission, against seed 0's exact table.

(a) HEADLINE: max-over-layers twin-minus-leak-reader gap at pre_color is at least 0.30

FAILED AS WRITTEN, AND IT WOULD HAVE FAILED AT SEED 0 TOO: +0.202 at seed 1, +0.233 at seed 0. The 0.496 baseline in the header is the twin's max against the leak-reader's L8 value. Scored at L8, the layer the project actually quotes, it is +0.496 -> +0.487 and replicates.

(b) ASYMMETRY: zero justifies_next features at pre_node, all seven layers, BOTH arms

CONFIRMED -- 0 at every layer in both arms, best F1 0.229 and 0.253 against nulls of 0.102 and 0.098. 6 dictionaries now, across three checkpoints and two seeds -- 42 layer-measurements, all zero.

(c) LAYER 0: best next_node F1 >= 0.99 at pre_color and zero next_node features at pre_node, in both arms (counts at pre_color deliberately NOT pre-registered)

CONFIRMED -- F1 1.0000 and 1.0000 at pre_color, 0 features at pre_node in both. Not pre-registering the counts was right: they went 23->50 and 24->50 while F1 held at 1.000, the same instability the solver showed.

(d) LOW CONFIDENCE: the leak-reader's justifies_next features stay LATE -- at L10/L12, with none at L6 or L8

CONFIRMED, and tighter than predicted: 13 at L10 and 0 at L12 (seed 0 had 13 and 2), zero at L6 and L8 in both seeds. The falsifier was any leaked feature at L6 or L8; it did not happen. This is the mechanism behind (a)'s two scorings.

3 of 4, with the miss on the one that mattered and for a reason that has nothing to do with the data. The lesson is narrow and worth keeping: write the pre-registered statistic as the expression you will compute, not as a sentence plus a remembered number.

What the second dictionary settles, and what it rewords

A contrast that held, and a sentence that has to name its layer

WHAT THIS BUYS

The contrast findings 19 and 21 are built on is no longer one dictionary per arm. At L8 it moves 0.009 between seeds against a gap of 0.487, and the pre_node asymmetry is now zero-for-zero in 6 dictionaries. Total justifies_next at pre_color: twin 69 -> 74, leak-reader 15 -> 13.

WHAT HAS TO BE REWORDED

Every place the project states the between-arm gap, it must name the layer. 'The leak-free model represents the forcing set and the leaked one does not' is false as a global claim -- the leak-reader has 13 forcing-set features at L10 at F1 0.519. The true claim is about timing: the twin has the forcing set inside the causal handoff window, the leak-reader only after it. Finding 29(e) already said this; this run makes it the headline.

WHAT IS STILL UNFIXED

best_f1 is a max over 256 predicates in every seed, so the multiple-comparisons ceiling is unchanged -- two seeds show the same ceiling gives the same answer twice, which is weaker than removing it. Two seeds is two, not a distribution. The SAE stays CORRELATIONAL: the feature-level ablation (finding 30) found no sensitivity at all, so what the model USES comes from the interventions in findings 27 and 28, not from here.

WHAT WOULD SETTLE THE TIMING CLAIM

Activation patching on the LEAK-READER, the same L6-L10 sweep finding 27 ran on the two leak-free models. If the timing story is right, its forcing fact should arrive late or not at all -- and it already flips at only 0.388, so there may be little to track. That is the test that would turn this correlational timing difference into a causal one.

Outputs under $SCRATCH/spin_gnn/sudoku_sae_kindsplit_2026-10-31_seed1/ (90-day purge; the summaries are small and the dictionaries are re-derivable from the checkpoints). Notebook: notebooks/sudoku_sae_arms_seed1_2026-10-31_results.ipynb. Job 50604600, repo at d0b4654, sudoku_sae.py at 10c38b0, seed 1.